

Current Multimedia, GUI and Media Formats — Detailed Lecture Notes

JKSSB Computer Awareness

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65.1 What multimedia means

Multimedia is the presentation of content that combines more than one medium in a single document or program. The media may be **text**, **graphics**, **audio**, **video**, and **animation**. Text and graphics are *static* media: they do not change with time. Audio, video, and animation are *time-based* media: their meaning depends on being played in the correct order and at the correct speed. Because a computer stores everything as binary data, every medium — sound, picture, or moving image — must first be captured and converted into numbers before it can be stored, compressed, and replayed.

Definition (Multimedia). *Content that combines two or more media types — such as text, graphics, audio, video, and animation — in a single digital presentation.*

Element	Meaning
Text	Readable characters; the base medium of almost every document.
Graphics	Still images, drawings, and photographs.
Audio	Sound, music, and speech.
Video	Moving pictures, usually accompanied by sound.
Animation	A sequence of images played fast enough to create motion.

65.2 Digitising media: sampling and quantization

An analogue signal, such as a sound wave or a scene's light, is *continuous*: it has a value at every instant. A computer, however, stores only discrete numbers. Converting a continuous signal into digital form is done by **analogue-to-digital conversion**, and it has two distinct steps, **sampling** and **quantization**.

Definition (Sampling). *The process of measuring a continuous signal at regular intervals and recording each measurement. The number of measurements taken per second is the **sampling rate**, measured in hertz (Hz).*

A **higher sampling rate** captures the signal more often, so the reproduction is more faithful, but the file becomes larger. A lower sampling rate saves space but loses detail. The sampling rate is the horizontal resolution of the digitised signal: how often we look at it.

Definition (Quantization). *The process of mapping each measured sample to one of a fixed set of discrete levels, because a computer can store only a limited number of distinct values per sample.*

The number of available levels is set by the **bit depth** of each sample. With a higher bit depth — more bits per sample — there are more levels, so each recorded value is closer to the true value;

the rounding error, called *quantization noise*, is smaller, and quality is better, but again the file is larger.

Example 65.2.1. *Think of drawing a smooth curve on graph paper. **Sampling** decides how many points across the page you mark — more points follow the curve more closely. **Quantization** decides how finely the vertical axis is ruled — more lines let you record each point's height more precisely. The two choices together determine how closely the digitised version matches the original, and how big the file becomes.*

The common exam confusion is to swap the two. Remember the one-line distinction: **sampling is how often we measure; quantization is how precisely each measurement is stored.**

65.3 Compression: lossy and lossless

Raw multimedia files are large, so they are almost always **compressed** — stored in a form that takes less space. There are two kinds of compression, and the difference is whether the original can be recovered exactly.

Definition (Lossless compression). *Compression that allows the original data to be reconstructed exactly. No information is discarded. PNG and GIF images and ZIP archives are lossless.*

Definition (Lossy compression). *Compression that discards some data to reduce the file size further, so the reconstruction is only an approximation of the original. JPEG images and MP3 audio are lossy.*

Lossless compression works by removing redundancy — repeated or predictable data — while keeping every value. Lossy compression goes further: it removes detail that a human eye or ear is unlikely to notice. Lossy files are usually much smaller, which is why lossy formats dominate the internet for photographs, music, and video. When exactness matters — for a diagram, screenshot, or archive — a lossless format is used instead.

65.4 Synchronization

A multimedia file often carries more than one **stream** at once: a video stream and an audio stream, for instance. **Synchronization** is the process of keeping those streams aligned in time so they play together correctly. If the streams drift out of step, the sound no longer matches the picture — the narrator's voice is heard before or after the lips move, which is called a loss of *sync*. Synchronization is what makes a video play with the sound in the right place at the right moment; it is a timing requirement peculiar to time-based media.

Definition (Synchronization). *The alignment in time of separate media streams — such as audio and video — so that they are presented together correctly during playback.*

65.5 The graphical user interface and its elements

A **GUI** is a **Graphical User Interface**: it lets the user work with pictures, icons, and menus instead of typing commands. A **CLI** is a **Command-Line Interface**: it accepts typed text commands. The GUI is easier for beginners and is the standard on modern desktops; the CLI is often faster for an expert and is still used for administration.

Definition (GUI (Graphical User Interface)). *A user interface in which the user interacts with the computer through visual elements such as windows, icons, menus, and a pointer, rather than by typing text commands.*

The four classic GUI elements are remembered by the short form **WIMP: Windows, Icons, Menus, Pointers**.

Element	Meaning
Window	A rectangular on-screen area that holds a program or a document.
Icon	A small picture that represents a file, folder, or program.
Menu	A list of commands that the software offers.
Pointer	The on-screen arrow, moved by the mouse or touchpad, used to select.

Definition (Dialogue box). *A small window that appears to ask the user for input or to confirm an action, such as Save As or Print. It usually contains text boxes, drop-down lists, check boxes, radio buttons, and OK/Cancel buttons.*

65.6 WYSIWYG

WYSIWYG stands for **What You See Is What You Get**. It means that what is shown on the screen is exactly how the document will look when printed — the same layout, fonts, spacing, and page breaks. A word processor such as MS Word works in WYSIWYG mode, so the page visible on the monitor matches the printed page.

Example 65.6.1. *While editing a document in MS Word, the text, margins, and page breaks on screen match the printed output. This is WYSIWYG. When the user opens the Print dialogue box and chooses options, those settings are entered in a dialogue box, not typed on the command line. The two ideas are often asked together: WYSIWYG describes the on-screen view; a dialogue box is one GUI element that collects input.*

65.7 Image formats: PNG, JPEG, and GIF

Three image formats appear most often in competitive-exam questions.

Format	Full form	Nature and typical use
PNG	Portable Network Graphics	Lossless; supports transparency; used for logos and web graphics.
JPEG	Joint Photographic Experts Group	Lossy; best for photographs and complex pictures.
GIF	Graphics Interchange Format	Lossless but limited to 256 colours; used for simple graphics and short animation.

JPEG is chosen for photographs because its lossy compression makes them much smaller with little visible loss. **PNG** is chosen when exact colours or transparency are needed, because it is lossless. **GIF** is chosen for simple graphics and small animations, but it can show only 256 colours, so it is unsuitable for photographs. **BMP** (Bitmap) is an older, largely uncompressed image format that produces large files and is not used for the web.

65.8 Audio and video formats: MP3 and MP4

MP3 stands for **MPEG-1 Audio Layer III** (also written MPEG Audio Layer III). It is a **lossy** audio format: it removes sound detail that the human ear is unlikely to notice, which is why an MP3 file is far smaller than an uncompressed one. Its small size made MP3 the standard for portable music and downloads. By contrast, **WAV** is an uncompressed audio format.

MP4 stands for **MPEG-4 Part 14**. It is a **container** format: a single MP4 file can hold video, audio, and subtitles together. It is widely used for streaming, downloads, and mobile video. A container such as MP4 holds the streams; the *codec* inside the container is what actually compresses the video and audio. **AVI** (Audio Video Interleave) is an older video container.

Definition (MPEG). *MPEG stands for Moving Picture Experts Group — the group that defines the MP3 audio layer and the MP4 container, among other standards.*

65.9 Frame rate, resolution, and bit rate

Video is a rapid sequence of still frames. The **frame rate** is the number of frames shown per second (fps); a higher frame rate produces smoother motion but a larger file. The **resolution** is the number of pixels in each frame (for example, 1920×1080); more pixels mean a sharper image and a larger file. The **bit rate** is the amount of data stored or sent per second; a higher bit rate keeps more detail but needs more space or bandwidth. These three settings, together with the choice of lossy or lossless compression, decide the final quality and size of a video or audio file.

Definition (Frame rate). *The number of still frames displayed per second in a video. A higher frame rate gives smoother motion at the cost of a larger file.*

Example 65.9.1. *A video at 30 frames per second with a high bit rate looks smooth and sharp but takes much more space than the same clip at a lower frame rate and bit rate. Streaming services lower the bit rate when the network is slow, trading quality for smooth playback — a daily example of the quality-versus-size trade-off that runs through the whole lesson.*

65.10 Plotter

A **plotter** is an **output** device that draws on paper. Unlike a printer, which lays down text and small images, a plotter produces **vector graphics**: lines and curves defined by coordinates rather than by a grid of dots. It is used for **large engineering drawings**, maps, blueprints, and banners, where continuous, accurate lines are needed. Early pen plotters moved a pen across the paper; modern inkjet plotters spray ink while the sheet moves. A plotter is always an output device, never an input device (PYQ-29, PYQ-08).

65.11 Impact versus non-impact printers (revisit)

The basic classification of printers reappears in later units, so keep it firmly in mind. An **impact printer** forms an image by striking the paper; a **non-impact printer** does so without any striking action.

Point	Impact	Non-impact
Action	Strikes the paper to print	Forms the image without striking
Examples	Dot-matrix, daisy-wheel, line printer	Inkjet, laser, thermal printer
Carbon copies	Possible (same strike hits several sheets)	Not possible
Noise	Comparatively noisy	Comparatively quiet

The quick test is the mechanism: if the device physically hits the paper, it is impact; if it does not, it is non-impact. Both kinds are output devices. A *dot-matrix* printer is impact; an *inkjet* and a *laser* printer are non-impact.

65.12 Multimedia hardware and applications

Playing and creating multimedia needs both hardware and software. On the hardware side, a **sound card** processes audio, **speakers** and **headphones** play it, and a **microphone** captures it. A **graphics card (GPU)** renders images and video, a **monitor or projector** displays them, and a **camera** or **webcam** captures still and moving pictures. Because audio and video files are large, multimedia also places heavy demands on **memory and storage**.

On the software side, editors for images, audio, and video are used to capture, edit, compress, and synchronize the media streams. The main application areas of multimedia are education (e-learning and tutorials), entertainment (music, games, and streaming films), business (presentations and advertising), and medicine (medical imaging and simulation).

Short form	Full form
GUI	Graphical User Interface
CLI	Command-Line Interface
WYSIWYG	What You See Is What You Get
WIMP	Windows, Icons, Menus, Pointers
PNG	Portable Network Graphics
JPEG	Joint Photographic Experts Group
GIF	Graphics Interchange Format
MP3	MPEG-1 Audio Layer III
MP4	MPEG-4 Part 14
MPEG	Moving Picture Experts Group
AVI	Audio Video Interleave
BMP	Bitmap

65.13 Exam retrieval and common confusions

Several distinctions prevent most errors on this topic.

1. **Sampling** is how often a signal is measured; **quantization** is how precisely each sample is stored. Do not swap them.
2. **Lossless** formats restore the original exactly (PNG, GIF); **lossy** formats discard data (JPEG, MP3, MP4).

3. **JPEG** is lossy and best for photographs; **PNG** is lossless and supports transparency; **GIF** is lossless and supports simple animation.
4. **MP3** is audio (MPEG-1 Audio Layer III); **MP4** is video (MPEG-4 Part 14). Do not swap the full forms.
5. **Synchronization** keeps audio and video aligned in time; it is not compression and not sampling.
6. **GUI** is Graphical User Interface; its elements are **Windows, Icons, Menus, Pointers** (WIMP).
7. **WYSIWYG** means the screen matches the print; a **dialogue box** is a small window that collects input.
8. A **plotter** is an output device for large drawings; an **impact** printer strikes the paper and a **non-impact** printer does not.

Example 65.13.1. *An item asks, “Which format is lossless and supports transparency?” Trace the formats: JPEG is lossy, MP3 is lossy audio, MP4 is lossy video, and GIF is lossless but limited to 256 colours without transparency support. The answer is PNG, the lossless format that supports transparency.*

Exercises

1. Define multimedia and name its five building blocks.
2. Distinguish sampling from quantization, and explain how each affects file size.
3. Differentiate lossy and lossless compression, giving one example of each.
4. What is synchronization in multimedia, and why is it needed?
5. Expand GUI, WYSIWYG, and WIMP.
6. State one suitable use each for PNG, JPEG, and GIF.
7. Expand MP3 and MP4, and state the medium each is used for.
8. Explain why a plotter is preferred over a printer for large engineering drawings.
9. Distinguish impact printers from non-impact printers with one example of each.

Solutions

1. Multimedia is content combining more than one medium in a single presentation. Its five building blocks are text, graphics, audio, video, and animation.
2. Sampling measures a continuous signal at regular intervals; a higher sampling rate captures more detail but enlarges the file. Quantization maps each sample to one of a fixed set of levels set by the bit depth; more levels improve quality but also enlarge the file.
3. Lossless compression restores the original exactly (for example PNG); lossy compression discards some data and is only an approximation (for example JPEG).

4. Synchronization is the alignment in time of separate media streams such as audio and video. It is needed so that sound and picture play together correctly rather than drifting apart.
5. GUI = Graphical User Interface; WYSIWYG = What You See Is What You Get; WIMP = Windows, Icons, Menus, Pointers.
6. PNG: logos and web graphics needing transparency or exact colours. JPEG: photographs and complex pictures. GIF: simple graphics and short animations.
7. MP3 = MPEG-1 Audio Layer III, used for audio (music and speech). MP4 = MPEG-4 Part 14, used for video (a container holding video, audio, and subtitles).
8. A plotter produces vector graphics — continuous lines defined by coordinates — on large sheets, which suits precise engineering drawings, maps, and blueprints better than a printer.
9. Impact printers strike the paper (for example dot-matrix); non-impact printers form the image without striking (for example inkjet or laser).